

Aspera Orchestrator Workflow Authoring Guide

IBM Aspera

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Chapter 1

Introduction

This guide explains how to write an Aspera Orchestrator workflow as a file, import it, publish it and run it, without the graphical designer and without exporting an existing workflow first.

It is written for people and for AI assistants. Each rule gives the exact element, attribute or value expected by Orchestrator.

The content is derived from the Orchestrator 4.1.6 source code. The examples in **Recipes** were imported, published and executed on Orchestrator 4.1.6.

Related documents:

- [Action Template Reference](#): attributes, inputs and outputs of every action plugin (generated from the plugin source).
- [Plugin Manual](#): functional description of each action plugin.
- [Plugin Development Guide](#): how to write an action plugin.

1.1 Terminology

Term	Meaning
Workflow	A named process definition, identified by an integer <code>id</code> and a <code>portable_id</code> (UUID).
Whiteboard	One revision of the design of a workflow: the graph, stored as XML in attribute <code>xml_export</code> .
Publish	Build the runnable workflow (steps, inputs, outputs, parameters, branches) from a whiteboard. Only a published revision runs.
Step	A node of the graph that executes an action.
Action plugin	The type of a step, for example <code>CustomRuby</code> . Also called step type or action type.
Action template	A configured instance of an action plugin: a database record holding the plugin settings (its attributes). Each step references one template by <code>Step_type</code> and <code>Action_id</code> .
Embedded template	A template that belongs to one workflow. Its name is <code>W<workflow id>__<step name></code> .
Global template	A template shared by workflows, identified by its name.
Parameter	A named value of the workflow. A runtime parameter can be set when starting the workflow.
Input, Output	Named values consumed and produced by a step, as declared by the action plugin.
Work order	One execution of a published workflow.
Work step	One execution of a step in a work order.

Chapter 2

Quick start

The following file is a complete workflow with one step that returns a message. It uses the **plain format**, which is the simplest to write.

```
---
- !ruby/object:Workflow
  attributes:
    name: hello
    run_as: SYSTEM
- !ruby/object:Whiteboard
  attributes:
    error_free: true
    xml_export: |
      <?xml version="1.0" encoding="UTF-8"?>
      <Workflow id="0">
        <Start x="100" y="50" id="start"><Parameters></Parameters></Start>
        <Error x="900" y="350"
↪ id="error"><Synch_factor>0</Synch_factor><Prerequisites></Prerequisites></Error>
        <End x="900" y="50" id="end"><Synch_factor>0</Synch_factor><Prerequisites><Prerequisite
↪ id="step1" status="Complete"/></Prerequisites></End>
        <Fail x="900" y="200"
↪ id="fail"><Synch_factor>0</Synch_factor><Prerequisites></Prerequisites></Fail>
        <Steps>
          <Step x="400" y="50" id="step1"><Action_id>1</Action_id><Step_name>Say
↪ hello</Step_name><Step_type>CustomRuby</Step_type><Synch_factor>0</Synch_factor><Prerequisites><Prerequisite
↪ id="start"
↪ status="true"/></Prerequisites><Parameters></Parameters><Inputs></Inputs><Outputs></Outputs></Step>
        </Steps>
        <Parameters></Parameters><Notes></Notes>
      </Workflow>
- CustomRuby__1: !ruby/object:CustomRuby
  attributes:
    name: W0__Say hello
    execute_code: "outputs['message'] = 'Hello'"
    typed_outputs: '{"message'=>'string'}"
- CustomRuby__1: []
- {}
- CustomRuby: 0.7.0
```

Import, publish and run it with the **IBM Aspera CLI** (**ascli**):

```
ascli orchestrator workflows import hello.yml # displays the new workflow, including its id,
↪ for example 239
ascli orchestrator workflows publish 239
ascli orchestrator workflows start 239 @json:'{}' @json:'{"step":"Say hello","variable":"message"}'
```

The last command waits for the work order to finish and displays **Hello**. The same operations with the REST API are described in **REST API**.

Chapter 3

Lifecycle

1. Write a workflow file (`.yaml`), or a package (`.wxf`) that also contains sub-workflows and remote nodes.
2. Import it: Orchestrator creates a new workflow, in draft state, with a new `id`. Each step's action template is created (or mapped to an existing global template).
3. Publish it: Orchestrator builds the runnable workflow from the last whiteboard revision.
4. Start it: Orchestrator creates a work order. Runtime parameters are given at this point.
5. Read the work order status and outputs.

Rules:

- Import and publish do not validate the graph. Publish only requires the whiteboard attribute `error_free` to be set. An invalid graph is published and fails at run time, or never ends. Use the [authoring checklist](#).
- An import through `import_workflow` always creates a new workflow. If the name is already used, Orchestrator appends a space and `(import_1)`, `(import_2)`, and so on. The `portable_id` of the file is ignored and a new one is generated.
- To add a revision to an existing workflow instead, use `import_with_constraints` with `add as revision`. See [Update an existing workflow](#).
- The API has no operation to delete a workflow: delete it in the user interface.

Chapter 4

Workflow file

4.1 File types

Extension	Content
<code>.yaml</code>	One workflow, as described in this guide.
<code>.wkt</code>	A gzipped tar archive containing a manifest file <code><name>.txt</code> and the files it lists. The manifest is a YAML list of file names: the first one is the main workflow (<code>.yaml</code>), the others are its dependencies: sub-workflows (<code>Workflow_*.yaml</code>) and remote nodes (<code>Remote_node*.node</code>). <code>export_workflow</code> produces this format when dependencies are exported.

The file name given to `import_workflow` must end with `yaml` or `wkt`.

4.2 Structure

A workflow file is a YAML sequence of exactly six elements, in this order:

#	Element	Content
1	Workflow	Workflow attributes: name, description, ... See Workflow attributes .
2	Whiteboard	The graph (<code>xml_export</code>) and its state. See Whiteboard attributes .
3	Action templates	A map: key <code><StepType>__<N></code> , value the template object. See Action templates .
4	Dependency list	A map: same keys as element 3, value the list of external entities used by the template. Empty lists for a workflow without dependencies.
5	Packed dependencies	A map of dependencies included in a <code>.wkt</code> package. <code>{}</code> for a <code>.yaml</code> file.
6	Plugin versions	A map: plugin name to version. Informational: not checked on import.

Rule for element 4 and 5: if element 4 lists dependencies and element 5 is empty, `import_workflow` does not import and returns the dependency list instead (see [import_workflow](#)). For a self-contained workflow, use empty lists in element 4.

The file can be written in two encodings. Both are accepted by the import.

4.3 Plain format

The plain format is the old (Orchestrator 4.0) format. It is the simplest to write, and the one used in this guide.

A file is in plain format when no line matches `concise_attributes:`. On import, Orchestrator converts it to the current format, line by line, and then loads it (`Compatibility::WorkflowYmlUpdater`). That converter is line-based, so the layout rules matter:

- An optional first line `---`.
- Each of the six elements starts with a dash and a space (`-` then space) at column 0.
 - Element 1 starts with `- !ruby/object:Workflow`.
 - Element 3 starts with `- <StepType>__<N>: !ruby/object:<StepType>` for the first template. Each next template starts with a line `<StepType>__<N>: !ruby/object:<StepType>` (any indentation). The separator must be exactly `: !ruby/object:`.
- A line `attributes:` is optional and ignored.
- The other lines of a section are its attributes, one `name: value` per line, with the same indentation within a section. Block scalars (`()`) are allowed, for example for `xml_export`.
- The class of a template is the part of its key before `__`.
- An unknown attribute name makes the import fail.
- In element 1, attributes with a null value are ignored.
- In element 3, attributes with a *blank* value are ignored: `null`, empty string, and also `false`. For a boolean attribute, write `1` or `0`, not `true` or `false`: `0` is kept, `false` is dropped (the database default then applies).
- Elements 4, 5 and 6 are required, even when empty.

On the server, the uploaded file is rewritten in the current format.

4.4 Rails format

This is the format produced by `export_workflow`: YAML serialization of Rails objects. Each attribute is an item of `concise_attributes:`

```
- !ruby/object:Workflow
concise_attributes:
- !ruby/object:ActiveModel::Attribute::FromDatabase
  name: name
  value_before_type_cast: My workflow
- !ruby/object:ActiveModel::Attribute::FromDatabase
  name: label
new_record: false
active_record_yaml_version: 2
```

An attribute without `value_before_type_cast` is null. In element 3, each template has the same form, under its key (`- CustomRuby__2: !ruby/object:CustomRuby`).

To read an exported workflow, look for the attribute names given in this guide. To write a workflow, prefer the plain format.

4.5 Workflow attributes

Columns of a workflow, and how `import_workflow` uses them:

Attribute	Type	On import
<code>name</code>	string	Used. Suffixed with a space and <code>(import_N)</code> if already used.
<code>description</code>	text	Used.
<code>run_as</code>	string	Used. One of <code>SYSTEM</code> , <code>CREATOR</code> , <code>OPERATOR</code> .
<code>label</code>	string	Used.
<code>tags</code>	text	Used. Stored as <code> tag1\\ tag2\\ </code> .

Attribute	Type	On import
templates	text	Used.
max_running	integer	Used. Maximum number of running work orders of this workflow.
cleanup_after_days	integer	Used.
purge_after_days	integer	Used.
user_lock, locked_by	boolean, integer	Used.
created_by	integer	Replaced by the importing user.
portable_id	string	Replaced by a new UUID.
source_revision	string	Replaced by <code><name>\ <revision></code> .
comments	text	Replaced by <code>Imported from file <path></code> .
id, folder_id, revision_id, generated_from, created_at, updated_at		Ignored. The folder is given by the import request.

The minimum is `name`, and `run_as` is recommended.

4.6 Whiteboard attributes

Attribute	Type	Rule
xml_export	text	Required. The graph. See Graph XML .
error_free	boolean	Required to publish. Publish returns <code>false</code> if it is null. Set it to <code>true</code> : it is not computed on import.
revision_id	integer	Optional. Defaults to 1.
created_by, workflow_id, revisions_log, id, created_at, updated_at		Ignored or replaced.

4.7 Action templates

Element 3 maps each key `<StepType>__<N>` to an action template.

- `<StepType>` is the plugin class name, for example `CustomRuby`, and must be the `Step_type` of the steps using it.
- `<N>` is a number unique within the file for that type, and must be the `Action_id` of the steps using it. On import, the template gets a new database id and the graph is updated: `Action_id` of steps, and `action_id` of `Map_to` and `Carry_thru` of type `output`.
- The value holds the template attributes: the plugin settings. The attribute names of each plugin are listed in the [Action Template Reference](#).
- Attribute `name` decides how the template is imported:
 - `W<any number>__<step name>`: embedded template. It is created for the new workflow and renamed `W<new workflow id>__<step name>`. Use this form, with `W0__` and the step name.
 - Any other name: global template. If a template of the same class and name exists and has the same attributes, it is reused. If it differs, a new template named `<name> (import_N)` is created. If none exists, it is created.
- Plugins that are not enabled on the server are enabled automatically by `import_workflow`. If that fails, the import returns the list of missing plugins.

How a template defines the inputs of its step is explained in [Inputs of a step](#).

Chapter 5

Graph XML

The whiteboard attribute `xml_export` holds the graph as XML.

5.1 Skeleton

```
<?xml version="1.0" encoding="UTF-8"?>
<Workflow id="0">
  <Start x="100" y="50" id="start"><Parameters></Parameters></Start>
  <Error x="900" y="350"
    id="error"><Synch_factor>0</Synch_factor><Prerequisites></Prerequisites></Error>
  <End x="900" y="50" id="end"><Synch_factor>0</Synch_factor><Prerequisites>...</Prerequisites></End>
  <Fail x="900" y="200" id="fail"><Synch_factor>0</Synch_factor><Prerequisites></Prerequisites></Fail>
  <Steps>
    <Step ...>...</Step>
  </Steps>
  <Parameters>...</Parameters>
  <Notes></Notes>
</Workflow>
```

General rules:

- The root element name is free, `Workflow` by convention. Its attribute `id` is replaced on import: use `0`.
- The elements `Start`, `End`, `Fail` and `Error` are required, exactly once each.
- At least one `Step` is required.
- `Parameters` (top level) and `Notes` are optional.
- Attribute `id` of every node is a string, unique in the graph, used to link nodes. The designer uses UUIDs; any unique string works, for example `start` or `step1`.
- Attributes `x` and `y` are the position of the node in the designer. They do not change execution.
- Usual XML escaping applies (`&`, `<`, `"`). Values of parameters and of inputs mapped to a value are also URL-decoded: `%XX` and `%uXXXX` sequences are decoded. Write `%25` for a literal `%` followed by two hexadecimal digits.

5.2 Terminal nodes

Element	Required children	Meaning
Start	none (<code>Parameters</code> optional)	Entry point. Its <code>Parameters</code> links declare the global parameters (see <code>Parameters</code>).
End	<code>Synch_factor</code> , <code>Prerequisites</code>	Reached: the work order is <code>Complete</code> .
Fail	<code>Synch_factor</code> , <code>Prerequisites</code>	Reached: the work order is <code>Failed</code> .

Element	Required children	Meaning
Error	Synch_factor, Prerequisites	Reached: the work order is Error.

Fail and Error are also reached implicitly: when a work step ends with status Failed (or Error) and no step is connected to it for that status, the work order goes to Fail (or Error). So only success paths need to be connected to End; failures do not need explicit links.

5.3 Step

Each Step element has attributes id, x, y, and these children:

Child	Required	Content
Action_id	yes	Number <N> of the template key <Step_type>__<N>.
Step_name	yes	Name of the step, unique in the workflow. Used to name outputs (<Step_name>: <output>) and in Map_to of type output.
Step_type	yes	Plugin class name, for example CustomRuby.
Synch_factor	yes	Number of prerequisites to meet. See Prerequisites and branches. The designer adds attributes x and y (position of the join symbol).
Prerequisites	yes (to be valid)	Links from previous nodes.
Inputs	yes (may be empty)	Inputs of the step. See Inputs.
Outputs	yes (may be empty)	Output settings and carry-throughs. See Outputs.
Parameters	no	Links to step parameters. See Parameters.
Version	no	Plugin version, informational.
Pre_processing	no	Ruby code run before the action. See Processing code.
Post_processing	no	Ruby code run after the action.
Timeout	no	Execution timeout in seconds. Default: timeout of the plugin, else the system default.
On_timeout	no	Status on timeout: Pass (Complete), Fail (Failed, the default), Error (Error), Raise (Error, and the work order timeout action runs).
Error_attempts	no	Maximum attempts when the step ends with Error and a branch with status Error leaves it.
Error_retryFor, Error_retryForUnit	no	Retry period: value multiplied by unit (seconds).
Error_retryDelay	no	Delay between attempts, in seconds.
Failed_attempts, Failed_retryFor, Failed_retryForUnit, Failed_retryDelay	no	Same, for status Failed.
comments	no	Description of the step.
weight	no	Integer, used for progress display.

Child	Required	Content
Journaling	no	Journal entry settings (Book, Event, Package, File, File_path, File_status, JournalStepMilestone, JournalProcessingTimeout). A missing journal book is created on import.

Element names are case-sensitive and must be written exactly as above. The parser ignores unknown elements: a misspelled optional element has no effect.

5.4 Prerequisites and branches

A `Prerequisite` element links a previous node to the current one:

```
<Prerequisites><Prerequisite id="check" status="Complete"/></Prerequisites>
```

Attribute	Content
id	id of the previous node.
status	Status of the previous node that activates the link. <code>true</code> for a link from <code>Start</code> . Otherwise the final status of the previous step: <code>Complete</code> , <code>Failed</code> or <code>Error</code> (compared case-insensitively).
router	Line style in the designer, for example <code>draw2d</code> . <code>ManhattanConnectionRouter</code> . Ignored by execution.

`Synch_factor` sets how many prerequisites must be met:

Value	Meaning
0	All prerequisites (AND).
N > 0	At least N prerequisites. 1 is OR: for example <code>End</code> after alternative branches.
negative	Documented in the source as "all but N", but computed as "number of prerequisites + N": never met. Do not use.

A link from `Start` always counts as met.

A step can have several outgoing links (several steps referencing it): steps whose prerequisites are met run in parallel.

To branch on a condition, make a step end with `Complete` or `Failed`, and connect different steps to each status. The `Filter` plugin does exactly that with a Ruby expression; a `CustomRuby` step can set `@status`. See [Recipe: branch on a status](#).

When alternative branches join, set `Synch_factor` of the join node (often `End`) to 1: with 0, it waits for all branches, and only one runs.

5.5 Inputs

```
<Inputs>
  <Input name="person"><Map_to type="parameter" name="who"/></Input>
  <Input name="text"><Map_to type="output" name="greeting" step_type="CustomRuby" action_id="1"
    ↪ step_name="Greet"/><Processing>input_value = input_value + '!'</Processing></Input>
</Inputs>
```

Attribute `name` of `Input` is the name of an input of the step, as declared by the action template (see [Inputs of a step](#)). An `Input` whose name is not declared by the template is silently ignored.

`Map_to` gives the source of the value:

type	Other attributes	Source
value	<code>name</code> : the literal value	A constant. For an input of type <code>array</code> , a value not starting with <code>[</code> is wrapped into a one-element array.
parameter	<code>name</code> : parameter name	A parameter (see Parameters).
global	<code>name</code> : parameter name	Same as <code>parameter</code> .
output	<code>name</code> : output name, <code>step_name</code> , <code>step_type</code> , <code>action_id</code>	An output of a previous step. <code>step_name</code> , <code>step_type</code> and <code>action_id</code> must be the <code>Step_name</code> , <code>Step_type</code> and <code>Action_id</code> of that step.

Optional child `Processing`: Ruby code that transforms the value (see [Processing code](#)).

Every required input of the template must have an `Input` element.

5.6 Outputs

The outputs of a step are those declared by its action template, plus `Step_information`. They all exist, whether or not the `Outputs` element lists them. Each output is a variable named `<Step_name>:<output name>`.

The `Outputs` element only adds settings:

```
<Outputs>
  <Output name="file_name"><Processing>output_value = output_value.upcase</Processing></Output>
  <Carry_thru type="output" name="file" step_type="ArrayFanout" action_id="33" step_name="Fan out"/>
</Outputs>
```

Element	Content
<code>Output</code>	Attribute <code>name</code> . Optional children: <code>Processing</code> (Ruby code transforming the value), <code>Global</code> (name of a global variable that also receives the value).
<code>Carry_thru</code>	Makes a variable available as an input of this step without the plugin declaring it. Attributes: <code>type</code> (<code>output</code> or <code>parameter</code>), <code>name</code> , and for <code>output</code> : <code>step_name</code> , <code>step_type</code> , <code>action_id</code> . Used for example after an <code>ArrayFanout</code> , to pass the current element through the steps of the fan-out.

Output `Step_information` is a `hash` with keys `step_id`, `step_name`, `step_status`, `step_status_details`, `attempts`, `processingTime`, `elapsedTime`, `completionTime`.

5.7 Parameters

A parameter is defined once in the top-level `Parameters` element, and linked to `Start` (global parameter) or to a step (step parameter):

```
<Start x="100" y="50" id="start"><Parameters><Parameter id="p_who"/></Parameters></Start>
...
<Parameters>
  <Parameter x="100" y="200" id="p_who">
```

```

<Name>who</Name><Value>world</Value><Value_type>string</Value_type>
<Runtime_override>true</Runtime_override><Runtime_optional>true</Runtime_optional>
</Parameter>
</Parameters>

```

Definition (`Parameter` with attributes `id`, `x`, `y`):

Child	Content
<code>Name</code>	Name, used in <code>Map_to</code> of type <code>parameter</code> and when starting the workflow.
<code>Value</code>	Value at design time, or default value of a runtime parameter. Use JSON for <code>array</code> and <code>hash</code> , for example <code>["/tmp/a", "/tmp/b"]</code> .
<code>Value_type</code>	One of the value types .
<code>Runtime_override</code>	<code>true</code> : the value can be given when starting the workflow. Default <code>false</code> .
<code>Runtime_optional</code>	<code>true</code> : a runtime value is optional, and <code>Value</code> is used when absent. Default <code>false</code> .
<code>Comments</code>	Description.

Link (`Parameter` with attributes `id`, and `router` for the designer) in `Start/Parameters` or `Step/Parameters`.

Rules:

- A parameter that is not linked to `Start` or to a step does not exist after publish: inputs mapped to it are silently not created.
- Runtime parameters are the inputs of the workflow: `workflow_inputs_spec` lists them, and `initiate` receives them as `external_parameters`.

5.8 Value types

Type	Content
<code>string</code>	Text.
<code>int</code>	Integer.
<code>float</code>	Floating point number.
<code>flag</code>	Boolean.
<code>date</code>	Date and time.
<code>array</code>	List (JSON array).
<code>hash</code>	Map (JSON object).
<code>object</code>	Any serializable value.
<code>pwd</code>	Password: hidden in the user interface, masked or encrypted on export.
<code>attachment</code>	File attachment.

5.9 Notes

`Notes` holds designer notes: `<Note id="..." note_id="..." x="..." y="..." collapsed="..."><Text>...</Text></Note>`. They do not change execution.

Chapter 6

Inputs of a step

The inputs of a step are declared by its action template, in method `inputs_spec` of the plugin: a map of required inputs and a map of optional inputs, each from name to **value type**. The [Action Template Reference](#) lists them for each plugin.

Many plugins derive inputs from some of their template attributes (`ActionTools#default_inputs_spec`). The [Action Template Reference](#) shows those attributes in column Input if blank. For these attributes:

- A text attribute left blank in the template becomes an input, so its value can come from the graph at run time.
 - Input name: the value of the plugin constant `VAR_<ATTRIBUTE>` if it exists, else the attribute name with its first letter in upper case (for example `file_path` gives `File_path`).
 - The input is optional if the plugin defines a constant `DEFAULT_<ATTRIBUTE>`, else required.
 - An attribute name ending with `_<type>` (for example `delay_int`) gives an input of that type, named without the suffix.
- A text attribute containing placeholders `<%= variable %>` creates a required input `variable` (type `string`). The placeholder is replaced by the input value at run time.
- A non-text attribute left null becomes an input of the corresponding type.

So the same plugin can have different inputs depending on its template. For example, `LocalFileWatcher` with `directory` left blank has a required input `directory`.

Plugins that let the template declare its inputs and outputs explicitly (for example `CustomRuby` and `Filter`) take them as a type map in text: JSON (`{ "person": "string" }`) or Ruby hash syntax (`{ 'person' => 'string' }`).

Before calling the action, Orchestrator adds these inputs: `OUTPUTS` (names and types of the outputs), `WORKORDER_ID` (the work step id), `ATTEMPT`, `RUNNING_AS`.

Chapter 7

Execution

7.1 Statuses

A work step ends with one of the following statuses, which are matched by the `status` of prerequisites:

Status	Meaning
Complete	Success.
Failed	Functional failure (for example a filter evaluated to false).
Error	Technical error (exception, timeout with <code>On_timeout</code> set to <code>Error</code>).

While running, a work step can also be `Inactive`, `In Progress`, `Paused`, `Canceling`, and so on. A work order is `Complete`, `Failed` or `Error` when it reaches `End`, `Fail` or `Error`, and `Canceled` when canceled.

7.2 Retries

When a step ends with `Error` (or `Failed`) and a branch with that status leaves it, the step can be retried before that branch is followed: up to `Error_attempts` (or `Failed_attempts`) attempts, within `Error_retryFor` x `Error_retryForUnit` seconds, waiting `Error_retryDelay` seconds between attempts.

7.3 Processing code

Ruby code can be attached to a step, an input or an output. It runs inside Orchestrator: it has full access to the server.

Location	Runs	Variables
Input/Processing	When the input value is read.	<code>input_name</code> (read-only), <code>input_value</code> (set it to change the value).
Step/Pre_processing	Before the action.	<code>@inputs</code> : map of all input values, modifiable. <code>@work_order_id</code> , <code>@state_id</code> (work step id).
Step/Post_processing	After the action.	<code>outputs</code> : map of output values, modifiable. <code>@status</code> , <code>@status_details</code> : set them to change the step result. <code>@work_order_id</code> , <code>@state_id</code> .
Output/Processing	When the output value is stored.	<code>output_name</code> (read-only), <code>output_value</code> (set it to change the value).

If processing code raises an exception, a warning is logged and execution continues with the unmodified value, unless the server option `strict_prepost_processing` is set, in which case the step ends with `Error`.

7.4 Triggers

Some plugins wait for an event instead of executing at once, for example `LocalFileWatcher`. They do not hold a worker while waiting. If the template enables continuous monitoring (attribute `allow_multiple` of `LocalFileWatcher`), each event starts a new work order that continues with the next steps, while the trigger keeps waiting.

Chapter 8

Main action plugins

The [Action Template Reference](#) lists the attributes, inputs and outputs of all plugins. This section details the plugins most useful to write workflows.

8.1 CustomRuby

Runs Ruby code. Inputs and outputs are declared in the template.

Attribute	Content
<code>execute_code</code>	Ruby code of the action.
<code>mandatory_inputs</code>	Required inputs: type map, for example <code>{'person' => 'string'}</code> .
<code>optional_inputs</code>	Optional inputs: type map.
<code>typed_outputs</code>	Outputs: type map, for example <code>{'greeting' => 'string'}</code> .
<code>inputs_spec_code</code>	Instead of <code>mandatory_inputs</code> and <code>optional_inputs</code> : Ruby code returning a map (required inputs) or an array of two maps (required, optional).
<code>outputs_spec_code</code>	Instead of <code>typed_outputs</code> : Ruby code returning the output type map.
<code>validate_inputs_code</code>	Ruby code returning <code>true</code> if the inputs are valid.
<code>use_code_from_github</code> , <code>github_*</code> , <code>source_control_type</code> , <code>use_cached</code> , <code>verify_ssl</code>	Get the code from a GitHub or Bitbucket repository instead of <code>execute_code</code> .

Contract of `execute_code`:

- Input values are in the map `@inputs`, for example `@inputs['person']`. An input can also be read as a method with its name, unless that name is also a method of the template: `name` returns the template name, not an input named `name`. Prefer `@inputs[...]`.
- Set outputs in the map `outputs` (or `@outputs`): `outputs['greeting'] = 'Hello'`. Only outputs declared in `typed_outputs` are stored.
- Optionally set `@status` to `Complete` (default), `Failed` or `Error`, and `@status_details` to a message. Any other status becomes `Error`.
- An exception ends the step with `Error`.

8.2 Filter

Evaluates a Ruby expression. The step ends with `Complete` if the result is true, `Failed` otherwise, and output `FilterResult` (`flag`) holds the result.

Attribute	Content
<code>command</code>	Ruby expression. <code><%= variable %></code> placeholders create required inputs.
<code>mandatory_inputs</code> , <code>optional_inputs</code>	Additional inputs: type maps.
<code>filter_outputs</code>	Additional outputs: type map.

8.3 LocalFileWatcher

Waits for files matching a pattern in a directory of the Orchestrator host. Output `FileName` is the full path of the file (`string`), or the list of paths (`array`) if `return_all` is set; `FileContent` holds the file content if `read_file` is set.

Attribute	Content
<code>directory</code> <code>file_name</code>	Directory to watch. Blank: becomes input <code>directory</code> . File name pattern (glob, or regular expression if use <code>regex_matching</code>). Blank: becomes input <code>File_name</code> .
<code>allow_multiple</code>	Continuous monitoring: start a new work order for each detection, and keep watching.
<code>remove_file</code>	Delete the file once detected.
<code>archive_directory</code> , <code>archive_with_timestamp</code> , <code>keep_timestamps</code>	Move the file to an archive directory once detected.
<code>cool_off</code>	Seconds a file must stay unchanged to be considered stable.
<code>polling_frequency</code>	Seconds between scans (default 5).
<code>partial_file</code>	Trigger on files still being written.
<code>return_all</code>	Return all matching files at once.
<code>trigger_type</code>	Persistence of past triggers: <code>by workorder group</code> (default), <code>by workflow</code> , <code>by action template</code> , <code>by step</code> , <code>none</code> .
<code>exclusions</code> , <code>fullpath_exclusions</code> , <code>ignore_zero_bytes</code> , <code>ignore_readlocked</code>	Filters.

Pitfall: with `remove_file` (and no archive directory) not set, a detected file stays in the directory. Past triggers are remembered per work order group by default, and each new work order is a new group: each new work order detects the same file again at once. Set `remove_file`, or `archive_directory`, so that a file is processed once.

8.4 ArrayFanout and Funnelin

`ArrayFanout` runs the next steps once per element of an array; `Funnelin` waits for all those executions and gathers their results.

- `ArrayFanout`: attribute `fanout_variable` is the name of the input (`array`), `fanned_as` the name of the output holding the current element, `variable_type` its type, `serialize` to run sequentially. Output `Fanout_Size` is the number of elements.
- `Funnelin`: attribute `funnel_variable` (and `funnel_type`) is the name of the input collected from each execution, `funneled_inputs` additional inputs (type map), `return_status` to add status outputs, `fail_if_failures` to fail if an execution failed. Output `Hash_by_state_id` maps each work step id to its value; each additional input `x` gives an output `x_hash`.

The steps between them use `Carry_thru` to pass the current element along.

Chapter 9

Recipes

The following workflows were imported, published and run on Orchestrator 4.1.6.

9.1 Recipe: minimal workflow

See [Quick start](#).

9.2 Recipe: runtime parameter and step chaining

Step `Greet` receives the runtime parameter `who`; step `Shout` receives the output of `Greet`, with input processing.

```
---
- !ruby/object:Workflow
  attributes:
    name: greet
    run_as: SYSTEM
- !ruby/object:Whiteboard
  attributes:
    error_free: true
    xml_export: |
      <?xml version="1.0" encoding="UTF-8"?>
      <Workflow id="0">
        <Start x="100" y="50" id="start"><Parameters><Parameter id="p_who"/></Parameters></Start>
        <Error x="900" y="350"
        ↪ id="error"><Synch_factor>0</Synch_factor><Prerequisites></Prerequisites></Error>
          <End x="900" y="50" id="end"><Synch_factor>0</Synch_factor><Prerequisites><Prerequisite
        ↪ id="shout" status="Complete"/></Prerequisites></End>
          <Fail x="900" y="200"
        ↪ id="fail"><Synch_factor>0</Synch_factor><Prerequisites></Prerequisites></Fail>
            <Steps>
              <Step x="300" y="50"
        ↪ id="greet"><Action_id>1</Action_id><Step_name>Greet</Step_name><Step_type>CustomRuby</Step_type><Synch_
                <Prerequisites><Prerequisite id="start" status="true"/></Prerequisites>
                <Parameters></Parameters>
                <Inputs><Input name="person"><Map_to type="parameter" name="who"/></Input></Inputs>
                <Outputs></Outputs></Step>
              <Step x="600" y="50"
        ↪ id="shout"><Action_id>2</Action_id><Step_name>Shout</Step_name><Step_type>CustomRuby</Step_type><Synch_
                <Prerequisites><Prerequisite id="greet" status="Complete"/></Prerequisites>
                <Parameters></Parameters>
                <Inputs><Input name="text"><Map_to type="output" name="greeting" step_type="CustomRuby"
        ↪ action_id="1" step_name="Greet"/><Processing>input_value = input_value +
                '!'</Processing></Input></Inputs>
                <Outputs></Outputs></Step>
            </Steps>
            <Parameters><Parameter x="100" y="200"
        ↪ id="p_who"><Name>who</Name><Value>world</Value><Value_type>string</Value_type><Runtime_override>true</R
```

```

    <Notes></Notes>
  </Workflow>
- CustomRuby__1: !ruby/object:CustomRuby
  attributes:
    name: W0__Greet
    mandatory_inputs: '{"person'=>'string'}'
    execute_code: "outputs['greeting'] = \"Hello #{@inputs['person']}\""
    typed_outputs: '{"greeting'=>'string'}'
CustomRuby__2: !ruby/object:CustomRuby
  attributes:
    name: W0__Shout
    mandatory_inputs: '{"text'=>'string'}'
    execute_code: "outputs['result'] = @inputs['text'].upcase"
    typed_outputs: '{"result'=>'string'}'
- CustomRuby__1: []
- CustomRuby__2: []
- {}
- CustomRuby: 0.7.0

```

```

ascli orchestrator workflows start <id> @json:'{"who":"Laurent"}'
↳ @json:'{"step":"Shout","variable":"result"}' # HELLO LAURENT!
ascli orchestrator workflows start <id> @json:'{}' @json:'{"step":"Shout","variable":"result"}'
↳ # HELLO WORLD!

```

9.3 Recipe: branch on a status

Step **Check** ends with **Failed** when the runtime parameter **value** is empty; one step is connected to each status; **End** waits for any of them (**Synch_factor** 1).

```

---
- !ruby/object:Workflow
  attributes:
    name: branch
    run_as: SYSTEM
- !ruby/object:Whiteboard
  attributes:
    error_free: true
    xml_export: |
      <?xml version="1.0" encoding="UTF-8"?>
      <Workflow id="0">
        <Start x="100" y="50" id="start"><Parameters><Parameter id="p_value"/></Parameters></Start>
        <Error x="1000" y="350"
↳ id="error"><Synch_factor>0</Synch_factor><Prerequisites></Prerequisites></Error>
        <End x="1000" y="50" id="end"><Synch_factor>1</Synch_factor><Prerequisites><Prerequisite
↳ id="on_ok" status="Complete"/><Prerequisite id="on_failed"
↳ status="Complete"/></Prerequisites></End>
        <Fail x="1000" y="200"
↳ id="fail"><Synch_factor>0</Synch_factor><Prerequisites></Prerequisites></Fail>
        <Steps>
        <Step x="300" y="50"
↳ id="check"><Action_id>1</Action_id><Step_name>Check</Step_name><Step_type>CustomRuby</Step_type><Synch_
        <Prerequisites><Prerequisite id="start"
↳ status="true"/></Prerequisites><Parameters></Parameters>
        <Inputs><Input name="value"><Map_to type="parameter"
↳ name="value"/></Input></Inputs><Outputs></Outputs></Step>
        <Step x="600" y="0" id="on_ok"><Action_id>2</Action_id><Step_name>On
↳ OK</Step_name><Step_type>CustomRuby</Step_type><Synch_factor>0</Synch_factor>
        <Prerequisites><Prerequisite id="check"
↳ status="Complete"/></Prerequisites><Parameters></Parameters><Inputs></Inputs><Outputs></Outputs></Step>
        <Step x="600" y="150" id="on_failed"><Action_id>3</Action_id><Step_name>On
↳ Failed</Step_name><Step_type>CustomRuby</Step_type><Synch_factor>0</Synch_factor>
        <Prerequisites><Prerequisite id="check"
↳ status="Failed"/></Prerequisites><Parameters></Parameters><Inputs></Inputs><Outputs></Outputs></Step>
        </Steps>
        <Parameters><Parameter x="100" y="200"
↳ id="p_value"><Name>value</Name><Value></Value><Value_type>string</Value_type><Runtime_override>true</Run

```

```

    <Notes></Notes>
  </Workflow>
- CustomRuby__1: !ruby/object:CustomRuby
  attributes:
    name: W0__Check
    optional_inputs: '{"value'=>'string'}'
    execute_code: "@status = @inputs['value'].to_s.empty? ? 'Failed' : 'Complete'"
CustomRuby__2: !ruby/object:CustomRuby
  attributes:
    name: W0__On OK
    execute_code: "outputs['result'] = 'value was provided'"
    typed_outputs: '{"result'=>'string'}"
CustomRuby__3: !ruby/object:CustomRuby
  attributes:
    name: W0__On Failed
    execute_code: "outputs['result'] = 'value was empty'"
    typed_outputs: '{"result'=>'string'}"
- CustomRuby__1: []
CustomRuby__2: []
CustomRuby__3: []
- {}
- CustomRuby: 0.7.0

```

With `{"value": "x"}`, only `On OK` runs; with `{}`, only `On Failed` runs. Both work orders end `Complete`.

9.4 Recipe: wait for a file

The workflow waits for a file matching `/tmp/ascli_watch_*.txt` on the Orchestrator host, deletes it, and returns its path.

```

---
- !ruby/object:Workflow
  attributes:
    name: wait_for_file
    run_as: SYSTEM
- !ruby/object:Whiteboard
  attributes:
    error_free: true
    xml_export: |
      <?xml version="1.0" encoding="UTF-8"?>
      <Workflow id="0">
        <Start x="100" y="50" id="start"><Parameters></Parameters></Start>
        <Error x="900" y="350"
⇒ id="error"><Synch_factor>0</Synch_factor><Prerequisites></Prerequisites></Error>
        <End x="900" y="50" id="end"><Synch_factor>0</Synch_factor><Prerequisites><Prerequisite
⇒ id="wait" status="Complete"/></Prerequisites></End>
        <Fail x="900" y="200"
⇒ id="fail"><Synch_factor>0</Synch_factor><Prerequisites></Prerequisites></Fail>
        <Steps>
          <Step x="400" y="50" id="wait"><Action_id>1</Action_id><Step_name>Wait for
⇒ file</Step_name><Step_type>LocalFileWatcher</Step_type><Synch_factor>0</Synch_factor>
          <Prerequisites><Prerequisite id="start"
⇒ status="true"/></Prerequisites><Parameters></Parameters><Inputs></Inputs><Outputs></Outputs></Step>
          </Steps>
        <Parameters></Parameters><Notes></Notes>
      </Workflow>
- LocalFileWatcher__1: !ruby/object:LocalFileWatcher
  attributes:
    name: W0__Wait for file
    directory: /tmp/
    file_name: ascli_watch_*.txt
    allow_multiple: 0
    remove_file: 1
    cool_off: 2
- LocalFileWatcher__1: []

```

```
- {}  
- LocalFileWatcher: 1.9.3
```

```
ascli orchestrator workflows start <id> @json:'{}' @json:'{"step":"Wait for  
↪ file","variable":"FileName"}'
```

The command returns when a matching file has been stable for `cool_off` seconds, for example `/tmp/ascli_watch_20261006_230726.txt`.

9.5 Update an existing workflow

`import_workflow` always creates a new workflow. To add a revision to an existing workflow, the file must be on the Orchestrator host, and `import_with_constraints` is called with `add as revision`:

1. Have the file on the server. `import_workflow` stores each uploaded file in `<run_dir>/workflows/import/<import_file_name>` (for example `/opt/aspera/orchestrator/var/run/orchestrator/workflows/import/`), but also imports it as a new workflow.
2. Call `import_with_constraints` with the path of the file and the id of the workflow to update (see `import_with_constraints`). The response is the workflow; its last revision is the imported one.
3. Publish the workflow: the new revision becomes the running one.

With `ascli`:

```
ascli orchestrator workflows import_with_constraints  
↪ @json:'{"filename":"/opt/aspera/orchestrator/var/run/orchestrator/workflows/import/my_workflow."  
↪ yml","add_as_revision":239}'  
ascli orchestrator workflows publish 239
```

Chapter 10

Authoring checklist

Import and publish do not check these rules. The designer checks most of them (`Whiteboard#parse_workflow`) before setting `error_free`.

File:

- Six elements, in order; elements 4, 5 and 6 present, even empty.
- Plain format: no line matching `concise_attributes`; booleans of templates written `1` and `0`.
- `error_free: true` in the whiteboard.
- Each template key `<StepType>__<N>` matches the `Step_type` and `Action_id` of its steps.
- Embedded templates named `W0__<step name>`.

Graph:

- `Start`, `End`, `Fail`, `Error` present once; at least one `Step`.
- `End`, `Fail`, `Error` and each `Step` have `Synch_factor`; each `Step` has `Inputs` and `Outputs` (may be empty).
- Each step has a name, unique in the workflow, and a type of an enabled plugin.
- Each step has at least one prerequisite (else it never runs: "Step is orphaned").
- Each step is the prerequisite of at least one node with status `Complete` or `true` (else: "Step does not have output").
- At least one prerequisite reaches `End`, `Fail` or `Error` (else: "Workflow does not terminate").
- Each `Map_to` of type `output` refers to an existing step, by `step_name`, `step_type` and `action_id`.
- Each `Map_to` of type `parameter` refers to a defined parameter, linked to `Start` or a step.
- Each required input of each template is mapped.
- Each parameter has a `Name` and a `Value_type`; a parameter that is not a runtime parameter, or that is an optional runtime parameter, has a `Value`.
- Join nodes after alternative branches have `Synch_factor` 1.
- Retry settings (`Error_*`, `Failed_*`) only on steps that have an outgoing branch with that status.

Chapter 11

REST API

Base URL: `https://<host>/aspera/orchestrator/api/`. Authentication: a JWT obtained with `POST api/login`, or query parameters `login` and `password` when JWT authentication is disabled. The response format is chosen with query parameter `format` (`xml`, the default, or `json`).

Routes: each endpoint below has a route with the given method. In addition, any endpoint can be called as `api/<endpoint>/<id>`, with any HTTP method, where `<id>` is the main identifier (workflow or work order).

Errors are returned with the exception message and stack trace in the response body.

11.1 import_workflow

`POST api/import_workflow`, body `multipart/form-data`:

Field	Required	Content
<code>import_file</code>	yes	The workflow file (file part).
<code>import_file_name</code>	yes	File name, ending with <code>yaml</code> or <code>wkf</code> . The upload is stored under this name in <code><run_dir>/workflows/import/</code> .
<code>folder_id</code>	no	Folder of the new workflow.
<code>link_to_subworkflows</code>	no	Link to existing sub-workflows instead of importing them.
<code>publish_workflow</code>	no	<code>true</code> or <code>1</code> : publish after import. The server option <code>publish_api_imported_workflow</code> sets the default.

Response: the new workflow as JSON (`{"workflow": {"id": ..., "name": ..., ...}}`), in both formats. Other responses with status 200:

- `{"plugins": [...], "missing_deps": [...]}`: plugins that could not be enabled; nothing is imported.
- `{"dependencies": ..., "file": ..., "workflow": ...}`: the file lists dependencies that are not packed; nothing is imported.

On error: status 500 with `{"error": ..., "message": ..., "details": ...}`.

11.2 export_workflow

`GET api/export_workflow/<id or portable id>` (no route without the id in the path)

Returns the workflow file (`.yaml`, Rails format). With parameter `export_with_dependencies`, returns a `.wkf` package with sub-workflows and remote nodes. Any non-empty value of `export_with_dependencies`, including `false`, enables it.

11.3 publish_workflow

POST `api/publish_workflow?id=<id or portable id>`, or `api/publish_workflow/<id or portable id>` with any method. The id can also be given in a JSON body `{"id": ...}`.

Publishes the last whiteboard revision. Response: `true` or `false` (JSON), or `<WorkflowPublish><Success>true</Success></WorkflowPublish>` (XML). `false` means the whiteboard has no `error_free` value.

11.4 find_constraints

GET `api/find_constraints?file=<path>[&wf_id=<workflow id>]`

Analyzes a workflow file present on the Orchestrator host and returns the list expected by `import_with_constraints`, with the possible choices for each conflict:

```
[{"filename": "<path>", {"add as revision": null}, {"subwf constraints": {}}, {"action template constraints": {}}, {"remote node constraints": {}}, {"missing plugins": null, "Auto-enable missing plugins?": null}]
```

It does not check that the file exists: a wrong path returns the same empty lists.

11.5 import_with_constraints

POST `api/import_with_constraints`, JSON body: an array of six objects, in this order:

```
[{"filename": "/opt/aspera/orchestrator/var/run/orchestrator/workflows/import/my_workflow.yml"}, {"add as revision": 239}, {"subwf constraints": {}}, {"action template constraints": {"Search Files": "Map to the existing Global Template"}}, {"remote node constraints": {"Baton": "Map to the existing Remote Node"}}, {"Auto-enable missing plugins?": "true"}]
```

Item	Content
filename	Path of the file on the Orchestrator host.
add as revision	Id of an existing workflow to add a revision to, or <code>null</code> to create a new workflow.
subwf constraints	Sub-workflow name to choice: <code>Map to the Existing Sub-Workflow</code> , <code>Create a new Sub-Workflow with a unique name</code> , <code>Add a new revision to the existing Sub-Workflow</code> .
action template constraints	Global template name to choice: <code>Map to the existing Global Template</code> , <code>Create a new Global Template with a different name</code> , <code>Override the existing Global Template with this new template</code> .
remote node constraints	Remote node name to choice: <code>Map to the existing Remote Node</code> , <code>Create a new Remote Node with a different name</code> , <code>Override the Existing Remote Node with this importing node</code> .
Auto-enable missing plugins?	<code>true</code> or <code>false</code> .

Response: the workflow as JSON. The workflow is not published.

11.6 initiate

POST `api/initiate?format=json`, JSON body `{"workflow_id": <id>, "external_parameters": {"<name>": <value>, ...}}`. Query parameters:

Parameter	Content
synchronous	true: wait for the end of the work order.
timeout	With synchronous: maximum wait, in seconds.
explicit_output_step, explicit_output_variable	With synchronous: return only this output.
explicit_output	Same, as <step name>:<output name>.
on_error_return_code, on_failed_return_code	HTTP status to return when the work order ends Error or Failed.
tags	Comma-separated tags of the work order.

11.7 Other workflow endpoints

Endpoint	Content
GET api/workflows_list	All workflows: id, name, portable id, published and latest revision.
GET api/workflow_inputs_spec/<id>	Runtime parameters: name, type, optional, default value.
GET api/workflow_outputs_spec/<id>	Outputs: <step name>:<output name>, type.
GET api/work_orders_list/<workflow id>	Work orders of a workflow.
GET api/work_order_status/<id>	Status of a work order.
GET api/work_order_output/<id>	Output values of a work order (XML: one variable element per output).
GET api/work_order_cancel/<id>	Cancel a work order.

11.8 ascli commands

Operation	Command
Import	ascli orchestrator workflows import <file>
Import with constraints, or as a revision	ascli orchestrator workflows import_with_constraints @json:'{"filename": "<path on server>","add_as_revision": <id>}'
Export	ascli orchestrator workflows export <id> --out.file=<file>
Export with dependencies (.wkf)	ascli orchestrator workflows export <id> @json:'{"dependencies":true}' --to-folder=<folder>
Publish	ascli orchestrator workflows publish <id>
Inputs, outputs	ascli orchestrator workflows inputs <id>, ascli orchestrator workflows outputs <id>
Start	ascli orchestrator workflows start <id> [<parameters>] [<execution>], where execution is {"synchronous":true} or {"step": "<step name>","variable": "<output name>"}
Work order	ascli orchestrator workorders status <id>, ascli orchestrator workorders output <id>, ascli orchestrator workorders cancel <id>

Chapter 12

Known issues

Observed in Orchestrator 4.1.6:

- A negative `Synch_factor` is never met (see [Prerequisites and branches](#)).
- `find_constraints` returns empty constraints for a file that does not exist.
- `export_workflow` exports dependencies for any non-empty value of `export_with_dependencies`, including `false`.
- `export_workflow` has no route without the workflow id in the path, although its source comment documents `?id=`.
- The `.wki` package produced by `export_workflow` with dependencies contains an empty copy of itself (the archive is built in the folder it packs).
- API errors return the exception stack trace, and often status 404 or 500 regardless of the cause.
- The example files of the product (`docs/workflow_examples`) contain `<Failed_retry_for_unit>`, while the parser reads `<Failed_retryForUnit>`: that setting is ignored.

End of document